Alpha on Trend-Following Beta, Revisited: A Sparse-Regularization Approach to Index Replication

Ulrik Simmelholt

October 11, 2026

Abstract

I replicate the SG Trend Index using the bottom-up and top-down decomposition of Jungle Rock (2024), with a different estimator for each component. The bottom-up component clusters 560 risk-parity trend strategies on their correlation structure alone, then fits a non-negative Lasso on the cluster medoids. The top-down component is a rolling Lasso on ten futures. Out of sample and net of transaction costs, their equal-weight combination earns 6.17% a year at 12.06% volatility, with a Sharpe ratio of 0.51 against 0.18 for the index and a correlation to the index of 0.78.

Robustness checks use the Probabilistic Sharpe Ratio (PSR) and repeat the backtest from every possible entry week in the out-of-sample period. The replication's Sharpe ratio exceeds the benchmark's at 98.54% of entry weeks, but a PSR of 0.89 falls short of the conventional 0.95 level, so the evidence of excess return is suggestive rather than conclusive.

Contents

1 Introduction **2**

2 Data **2**

3 Index replication **3**

3.1 Bottom-Up Replication 3

3.1.1 Strategy universe 3

3.1.2 Hierarchical Clustering 3

3.1.3 LASSO Regression 4

3.2 Top-Down Replication 5

3.3 Validation methodology 5

3.4 Results 5

4 Robustness **6**

4.1 Statistical significance 6

4.2 Robustness to entry timing 7

5 Discussion and Limitations **7**
5.1 Selection bias and the Deflated Sharpe Ratio 7
5.2 Split sensitivity 8

6 Conclusion **8**

1 Introduction

This white paper builds on Jungle Rock (2024), who replicates the Société Générale Trend Index by decomposing it into a bottom-up and a top-down component. That framework is closely related to the broader time-series momentum literature, most notably Hurst et al. (2013), which documents that past returns contain information about future returns.

Where I diverge from Jungle Rock (2024) is in how each of the two components are estimated. For the bottom-up component, rather than the iterative backward-elimination procedure used in the original paper, I adopt a two-stage estimation strategy that separates into an *unsupervised* and *supervised* learning element: candidate trend strategies are first grouped by their mutual correlation structure, after which a non-negative Lasso regression is fit only on the resulting, decorrelated representatives.

For the top-down component, I replace the rolling ElasticNet estimator of the original paper with a pure L_1 specification. This choice is motivated by the greater sparsity induced by the L_1 penalty, which translates into a more efficient trading implementation.

2 Data

Following Jungle Rock (2024), I use the same 10-asset futures universe as the original paper, listed in Table 1 and spanning equities, commodities, energy, fixed income, and FX.

Table 1: Asset universe and classification

Asset	Class
ES1 Index	Equities
SB1 Comdty	Commodities
HG1 Comdty	Commodities
GC1 Comdty	Commodities
C 1 Comdty	Commodities
CL1 Comdty	Energy
TY1 Comdty	Fixed Income
GBPUSD Curncy	FX
EURUSD Curncy	FX
USDJPY Curncy	FX

The dataset covers daily observations from 1 January 2002 to 18 September 2026, split chronologically 80%/20% into a training and a held-out test period thereafter. Returns are computed as simple daily returns, $r_t = (P_t - P_{t-1})/P_{t-1}$, rather than log returns, since the WTI crude oil contract (CL1) settled at −\$37.63 on 20 April 2020.

3 Index replication

3.1 Bottom-Up Replication

3.1.1 Strategy universe

Following Jungle Rock (2024), I construct the bottom-up replication from a large collection of trend-following strategies, each defined by a tuple (M, T, N, assets): a basket size M, a lookback T and a rebalancing interval N (both in trading days), and a randomly drawn subset of M assets from the universe in Table 1.

For a given asset and lookback T, the trend signal follows the measure of Tzotchev et al. (2015),

$$S_T = 2\,\Phi\left(\sqrt{T}\,\frac{\bar{r}_T}{\hat{\sigma}_T}\right) - 1, \tag{1}$$

where $\bar{r}_T$ and $\hat{\sigma}_T$ are the mean and standard deviation of daily returns over the trailing T-day window, and $\Phi(\cdot)$ is the standard normal cumulative distribution function. Only the sign of S_T is used, giving a long or short position with no flat state. Position sizing follows a two-stage risk-parity scheme, matching the original paper: a fixed annualized volatility target of 15% is first allocated equally across the distinct asset classes in the strategy's basket. Positions are re-signed and resized every N trading days, using only data available up to the day preceding the rebalance, and held fixed until the next rebalance.

I follow the original paper's parameter grid for N and the basket size M, and extend the lookback grid to four values rather than three: $T \in \{21, 63, 126, 252\}$ trading days, $N \in \{10, 25\}$ trading days, and $M \in \{3, 6, 9\}$ assets. For each value of M, asset subsets are sampled randomly rather than enumerated exhaustively: all $\binom{10}{9} = 10$ possible subsets are used in full for $M = 9$, while for $M = 3$ and $M = 6$, 30 subsets are drawn without replacement out of the 120 and 210 possible combinations, respectively. This yields a total universe of 560 candidate strategies.

3.1.2 Hierarchical Clustering

With a candidate universe of several hundred trend strategies many of them exhibit severe pairwise correlation differing from one another only by the substitution of a single asset. Before fitting any model against the target index, I therefore first reduce this redundancy directly through unsupervised learning.

I deliberately steer away from methods for dimensionality reduction such as PCA on economic-interpretability grounds since there is no mechanism to recover which specific basket size, lookback, or rebalancing frequency is actually driving the index's behaviour. I want to preserve a direct, nameable link between the final replicating portfolio and the economic parameters (M, T, N, assets) that define each strategy.

I instead use agglomerative hierarchical clustering on the correlation structure of the strategy universe, performed independently within each basket size M. Each pair is assigned a distance $1 - |\rho|$, where ρ is the pairwise correlation of their return series, and clusters are merged using average linkage. For each M, the number of clusters k is chosen to maximize the silhouette score,

so the degree of reduction is data-driven rather than fixed in advance. Within each cluster, I retain the member closest, on average, to the rest of its cluster as the sole representative, thus ensuring every surviving strategy remains a genuine, nameable (M, T, N, assets) trend system rather than a synthetic combination. Figure 1 shows this for $M = 9$, where strategies merge at very low distances, confirming the expected redundancy.

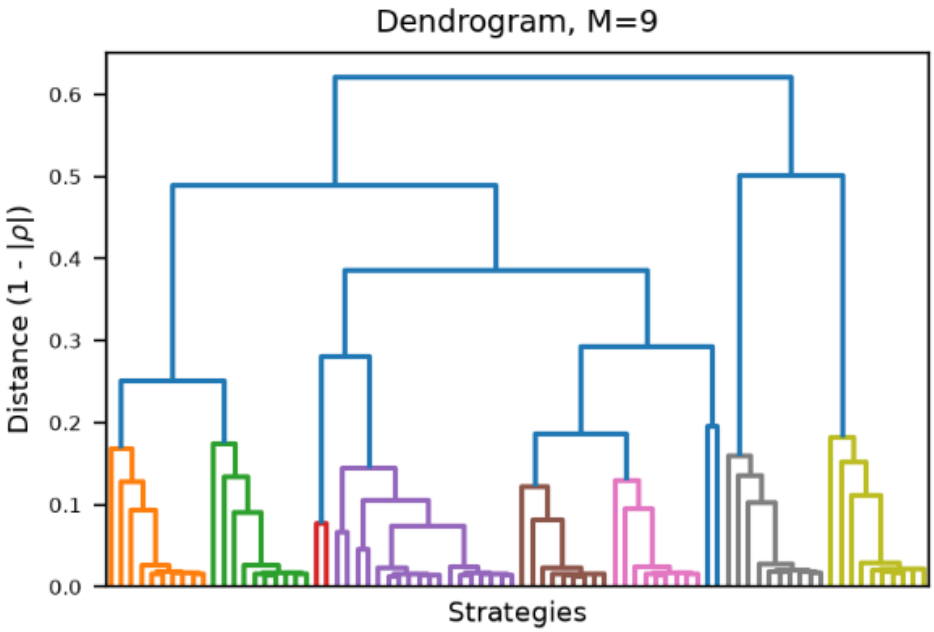


Figure 1: Dendrogram for the $M = 9$ group, coloured by the silhouette-selected clustering.

3.1.3 LASSO Regression

Having reduced each basket-size group M to its set of cluster representatives, I fit a non-negative Lasso regression of the target index on the surviving strategies,

$$\hat{\beta} = \arg\min_{\beta \geq 0} \quad \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1, \tag{2}$$

where X contains the daily return series of the surviving, decorrelated strategies and y is the target index's own daily return. The non-negativity constraint mirrors the original paper's own restriction to a positive linear combination of trend systems: a selected strategy enters the replicating portfolio as a long position in that strategy, and is never permitted to flip sign. The penalty weight λ is selected by time-series cross-validation on the training sample alone, using expanding windows so that no fold is ever validated on data preceding its own training set.

Because the preceding clustering stage already resolves the bulk of the collinearity present in the original strategy universe, the role of the Lasso penalty here is less about guarding against an ill-conditioned design matrix and more about two further tasks: discarding representatives that, despite surviving clustering on correlation grounds alone, do not materially improve the fit to the target index, and keeping the final replicating portfolio parsimonious, which matters directly for the transaction costs.

3.2 Top-Down Replication

Following Jungle Rock (2024), the top-down component approximates the target index as a direct linear combination of the 10-asset universe,

$$T(t) \approx \sum_{i=1}^{10} w_i(t)\, A_i(t), \tag{3}$$

with weights re-estimated periodically on a trailing window of recent returns, in order to track a relationship that is expected to drift over time rather than remain fixed across the full sample.

The original paper selects, at each refit, whichever of several linear models performs best by cross-validation, finding ElasticNet selected most often. In my implementation, ElasticNet and Lasso produce similar out-of-sample results, so for economic reasons, namely the need for trading efficiency given the short rolling window used here, I adopt the sparser Lasso specification throughout.

Following a short rolling-window sensitivity check, the estimation window and refit frequency are themselves selected by expanding-window cross-validation on the training sample, rather than fixed at the original paper's values, since the relationship between the assets and the index is found to vary meaningfully with both parameters.

3.3 Validation methodology

All reported results follow a strict separation between model development and evaluation. The sample is split chronologically into a training period and a genuinely held-out test period (Section 2).

Within the training period, hyperparameters are selected via expanding-window cross-validation rather than a single arbitrary train/validation split. These include the regularization strength in the Lasso regressions, the number of clusters in the bottom-up stage, and the estimation window and refit frequency in the top-down stage. Each fold's validation set follows strictly after its own training set, so that no fold is ever validated on data that precedes it; this is a deliberate departure from standard k-fold cross-validation, which permits a model to be validated on data that occurred before some of its training data, an implicit look-ahead that is particularly inappropriate for the fast-moving, short-lived relationships documented in Section 3.2.

Transaction costs are incorporated throughout at a flat rate of 2 basis points per unit of portfolio turnover per rebalance, following Jungle Rock (2024). Turnover is computed at the strategy level as the sum of absolute changes in position weight between consecutive rebalances; all net-of-cost figures reported in this paper deduct this cost directly from that rebalance's return.

3.4 Results

Figure 3 shows shows the growth of one dollar invested in the EW Replication over the out-of-sample period, net of transaction costs, against the SG Trend Index. The EW Replication earns an annualized return of 6.17% at an annualized volatility of 12.06%, a Sharpe ratio of 0.51, and

a maximum drawdown of 27.06%. The benchmark earns 4.01% at a volatility of 21.72%, with a Sharpe ratio of 0.18 and a maximum drawdown of 43.71%. The replication therefore delivers a higher return at roughly 55% of the benchmark's volatility, with a daily correlation to the index of 0.78.

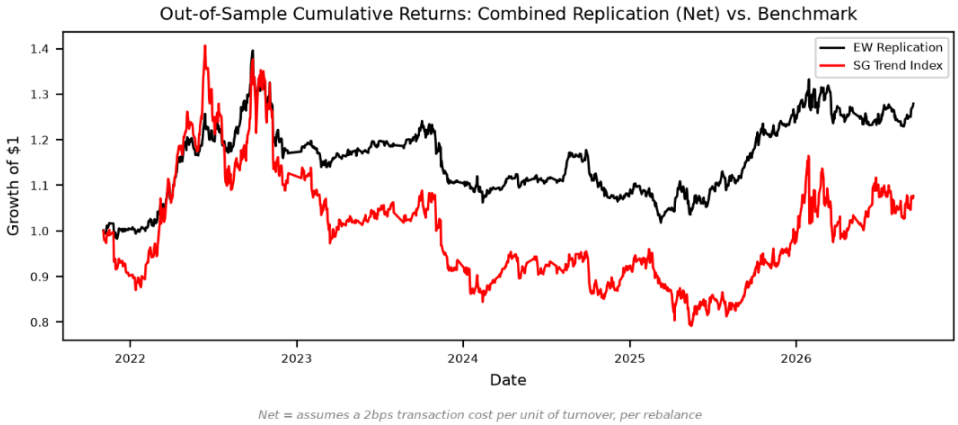


Figure 2: Cummulative out-of-sample return

4 Robustness

4.1 Statistical significance

To assess whether the reported out-of-sample performance reflects genuine skill rather than noise, I report the *Probabilistic Sharpe Ratio* (PSR) of Bailey and López de Prado (2012). PSR gives the probability that a strategy's true Sharpe ratio exceeds a benchmark level SR^* (here, zero):

$$\widehat{PSR}(SR^*) = Z\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right), \tag{4}$$

where $\widehat{SR}$ is the per-period Sharpe ratio, T the number of observations, $\hat{\gamma}_3$ and $\hat{\gamma}_4$ the sample skewness and kurtosis, and $Z(\cdot)$ the standard Normal CDF. A shorter sample, more negative skew, or heavier tails all widen the denominator, lowering confidence in a given $\widehat{SR}$ for a fixed point estimate. A PSR above 0.95 is conventionally taken as statistically significant.

Table 2 reports the PSR for the EW Replication and the benchmark index. The EW Replication's PSR of 0.89 exceeds the benchmark's 0.68, but falls short of the 0.95 level, so the evidence of a positive true Sharpe ratio is suggestive rather than conclusive. This is largely a consequence of the limited length of the out-of-sample period and the fat-tailed, negatively skewed return distribution, both of which reduce the precision of the Sharpe ratio estimate. Note also that PSR is a single-strategy test: it does not adjust for the model variants evaluated during development.

	EW Replication	SG Trend Index
Daily Sharpe Ratio	0.0322	0.0132
Annualized Sharpe Ratio	0.51	0.21
Skewness	−0.89	−1.13
Kurtosis	8.61	11.41
Probabilistic Sharpe Ratio (vs. $SR = 0$)	0.89	0.68

Table 2: Significance of out-of-sample performance: EW Replication vs. benchmark

4.2 Robustness to entry timing

A single train/test split reports performance for one specific entry date into the out-of-sample period, raising a natural concern: does the reported result depend on having started the backtest at a favourable moment? To address this, I compute the Sharpe ratio an investor would have realized by entering at *every* possible week within the out-of-sample period and holding through to its end, rather than reporting performance from a single fixed starting point.

For entry weeks near the very end of the sample, this calculation becomes numerically unstable: with only a handful of weeks remaining, the realized annualized volatility collapses toward zero, causing the resulting Sharpe ratio to explode upward. I therefore exclude the final 50 entry weeks from the analysis, retaining only entry points for which the Sharpe ratio is estimated over a sufficiently long remaining holding period to be meaningful.

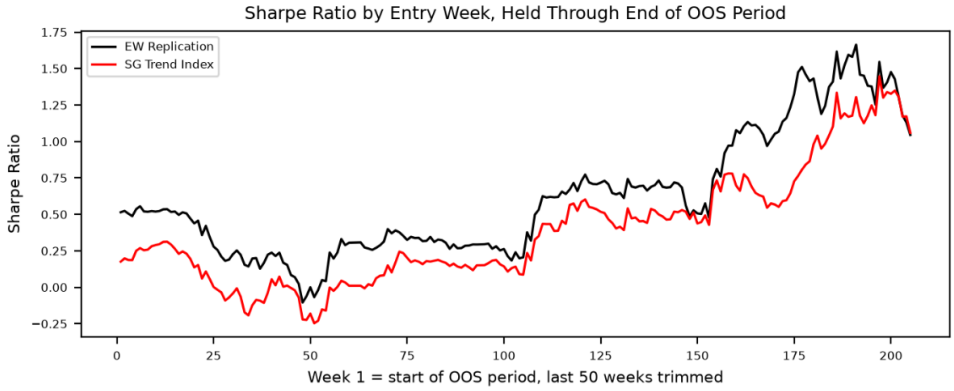


Figure 3: Sharpe ratio by entry week, held through the end of the OOS period.

Figure 3 plots the resulting Sharpe ratio for the EW Replication and the benchmark index as a function of entry week. Across the remaining entry points, the EW Replication's Sharpe ratio exceeds the benchmark's in 98.54% of cases, suggesting that the result is not driven by the particular week selected to begin the backtest.

5 Discussion and Limitations

5.1 Selection bias and the Deflated Sharpe Ratio

The PSR from 4.1 is conditional on several modelling choices made with knowledge of out-of-sample results, including the choice of Lasso over ElasticNet, the basket-size grid, and the

train/test split ratio itself. The Deflated Sharpe Ratio (Bailey and López de Prado, 2014) corrects for this by evaluating PSR not against zero, but against the maximum Sharpe ratio expected by chance across the trials conducted,

$$\widehat{DSR} = \widehat{PSR}\big(\widehat{SR}_0\big), \qquad \widehat{SR}_0 = \sqrt{V\big[\{\widehat{SR}_n\}\big]}\left((1-\gamma)\,Z^{-1}\left[1-\frac{1}{N}\right] + \gamma\,Z^{-1}\left[1-\frac{1}{N}e^{-1}\right]\right), \tag{5}$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant, $V[\{\widehat{SR}_n\}]$ is the variance of the Sharpe ratios across trials, and $\hat{N} = \hat{\rho} + (1-\hat{\rho})M$ is the effective number of independent trials implied by M trials with average pairwise correlation $\hat{\rho}$.

A rigorous application requires a complete, pre-specified record of all variants evaluated, which I did not maintain. As an illustration only, I apply it to the three variants still available at the time of writing (the bottom-up replication at $M \in \{3, 6, 9\}$, the wide-M bottom-up variant, and the top-down replication). Their average pairwise correlation of 0.73 implies $\hat{N} \approx 1.5$, giving a daily threshold $SR_0 = 0.0021$ and a DSR of 0.84 for the EW Replication, somewhat below its PSR of 0.89. The two are close because the three variants are near-identical in performance, so the expected maximum by chance is small. Both fall short of the conventional 0.95 level.

5.2 Split sensitivity

The 80/20 train/test split is a somewhat arbitrary choice, and reported performance depends on it. Splits whose test period includes the COVID-19 episode of early 2020 produce markedly stronger trend-following results than those that exclude it, and the split used here was chosen in part to avoid that. To limit the dependence on any single evaluation window, I report the Sharpe ratio for every possible weekly entry point within the out-of-sample period, held through to its end (Section 4.2), rather than relying only on performance measured from the first test day. This somewhat mitigates the choice of start date within the test period, but it does not remove the dependence on the train/test boundary itself, since the models are fitted once on the training sample.

6 Conclusion

I replicate the SG Trend Index with a bottom-up and a top-down component, following Jungle Rock (2024) but departing from it in how each is estimated. The bottom-up component clusters a universe of trend strategies on their correlation structure alone and fits a non-negative Lasso on the cluster medoids. The top-down component is a rolling Lasso on the ten underlying assets. Their equal-weight combination, the EW Replication, earns 6.17% a year at a volatility of 12.06% out of sample and net of transaction costs. Its Sharpe ratio is 0.51 against 0.18 for the benchmark, with a correlation to the index of 0.78, and it exceeds the benchmark's Sharpe ratio at 98.54% of the entry weeks tested, so the result does not depend on when the backtest starts.

These results carry caveats. A PSR of 0.89 and a DSR of 0.84 both fall short of the conventional 0.95 level.

References

David H. Bailey and Marcos López de Prado. The sharpe ratio efficient frontier. *Journal of Risk*, 15(2):3–44, 2012.

David H. Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

Brian Hurst, Yao Hua Ooi, and Lasse Heje Pedersen. Demystifying managed futures. *Journal of Investment Management*, 11(3):42–58, 2013.

Jungle Rock. Alpha on trend-following beta: A case study of the SocGen trend index. Technical report, Jungle Rock, December 2024. Working paper.

D. Tzotchev, M. Kolanovic, A. Lau, R. T. Krishnamachari, and D. Silvestrini. Designing robust trend-following systems: Behind the scenes of trend-following. Technical report, J.P. Morgan, 2015.